Brain Economy-Aligned Graph Transformer

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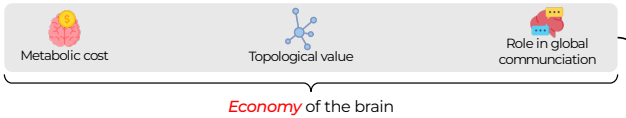
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Motivation

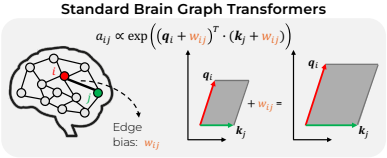
The brain connectivity is governed by an economy trade-off: neural connections balance their anatomical wiring cost against their contribution to efficient network communication [1,2]. Yet existing brain graph learning architectures do not explicitly incorporate this cost-value trade-off into their attention mechanism.



We formulate three hypotheses:

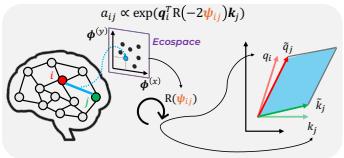
- ***H1*** **Edge-level economy**
  The cost and communication value of a neural pathway are properties of the connections, rather than its individual brain regions
- ***H2*** **Economy-aligned topology**
  Weighting connectivity by anatomical cost provides more informative topological representations than raw connectivity alone
- ***H3*** **Economy-aware attention**
  Embedding *brain economy* directly into the query-key interaction through asymmetric rotations is more effective than incorporating it as an external feature

Standard Brain Graph Transformers



Edge added symmetrically. Feature similarity is edge-agnostic.

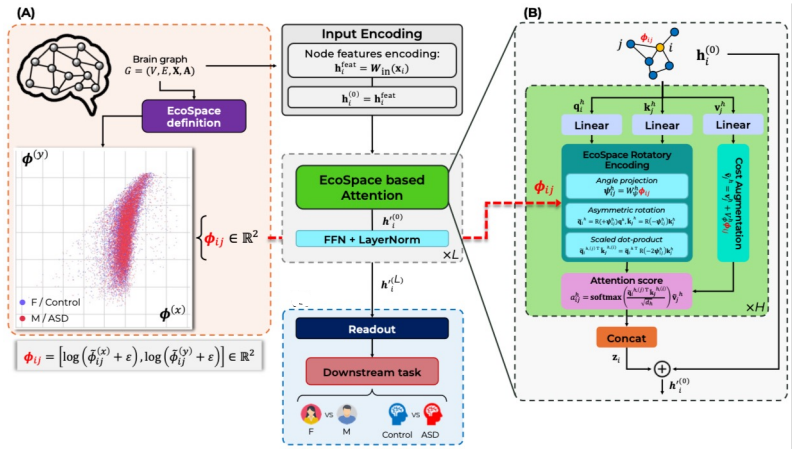
BGT-ESRE (ours)



Edge geometry rotates the query-key subspace. Feature similarity is instrinsically economy-dependent.

Methods

We introduced the **Brain Graph Transformer with Ecospace Rotary Encoding (BGT-ESRE)**, a novel graph transformer that embeds the brain connectivity economy into its attention mechanism, learning economy-aligned representations of brain communication.



1. Economy-aligned brain graph (A)
We designed an **economy-aligned adjacency matrix** by penalizing brain connectivity strength according to anatomical distance, incorporating the wiring cost-value trade-off:

$$A_{ij}^{\text{eco}} = BC_{ij} \cdot e^{-\lambda d_{ij}} A_{ij}$$

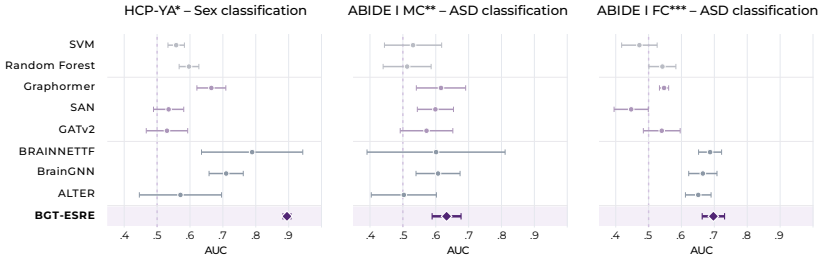
2. Ecospace construction (A)
Each brain connection is mapped into a 2D **ecospace**, defined by two graph communication measures computed on the economy-aligned adjacency matrix.

3. Ecospace Rotary Encoding (ESRE) (B)
We introduced an **edge-conditioned rotary attention mechanism**, where ecospace coordinates generate head-specific angles that rotate *query* and *key* vectors in opposite directions, making attention explicitly dependent on brain connection economy [3]. A gated projection additionally conditions *value* representations.

Key innovations
Brain connection economy directly modulates the query-key interaction, rather than being introduced as an external feature or additive attention bias.

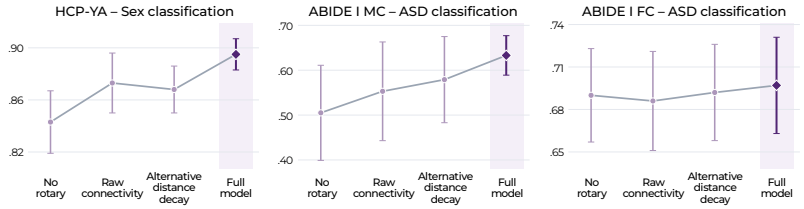
Results

a. Performances comparison
BGT-ESRE achieved the highest mean AUC across all three datasets, outperforming eight baseline methods:



*Human Connectome Project Young Adults (HCP-YA), resting-state fMRI, functional connectivity (FC) as input, 748 healthy participants, 405 female, 343 male.
**Autism Brain Imaging Data Exchange I (ABIDE I), resting-state sMRI, morphological connectivity (MC) as input, 341 participants, 155 neurotypical controls, 186 patients with autism spectrum disorder (ASD)
***Autism Brain Imaging Data Exchange I (ABIDE I), resting-state fMRI, functional connectivity (FC) as input, 884 participants, 476 neurotypical controls, 408 patients with autism spectrum disorder (ASD)

b. Ablation study
Both economy-aligned topology and rotary encoding contribute to performance. Replacing the economy-aligned graph with raw connectivity, removing rotary encoding, or modifying the distance-decay parameter consistently reduced mean AUC.



c. Ecospace attention analysis
Learned attention exhibits task-dependent ecospace patterns. Group-related differences in attention weights are associated with distinct communication properties, including diffusion efficiency, communicability and betweenness centrality.

Discussion

- **Economy-aware learning:** embedding brain connectivity economy directly into attention provides a useful inductive bias for brain graph classification.
- **Segregation-integration balance:** the highest-performing ecospace configurations combine segregation-related and integration-related topological measures, suggesting that these complementary dimensions are important for characterizing brain communication.
- **Neurobiological interpretation:** learned attentions reveals task-dependent patterns of economy-aligned communication, providing hypothesis-generating insights into brain network organization.

References
[1] Bullmore, Ed, and Olaf Sporns. "The economy of brain network organization." Nature reviews neuroscience 13.5 (2012): 336-349.
[2] Seguin, Caio, Olaf Sporns, and Andrew Zalesky. "Brain network communication: concepts, models and applications." Nature reviews neuroscience 24.9 (2023): 557-574.
[3] Su, Jianlin, et al. "Roformer: Enhanced transformer with rotary position embedding." Neurocomputing 568 (2024): 127063.

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